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RESEARCH ARTICLE

Daily Load Forecasting Using Explainable AI-Driven Method With Integrated Error Correlation

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ABSTRACT This paper presents a novel, explainable, and weather-independent day-ahead load forecasting (DALF) method. The proposed methodology integrates calendar-based segmentation using Classification and Regression Trees (CART), hybrid modeling with Multiple Linear Regression (MLR) and Gaussian Process Regression (GPR), and post-hoc error correction via Generalized Least Squares with Autoregressive Residuals (GLSAR). To enhance interpretability, Shapley Additive exPlanations (SHAP) are employed to quantify the influence of lagged load features and calendar variables across segmented forecasts. The framework is validated using half-hourly data from Thailand's Electricity Generating Authority (EGAT) and cross-tested against France's national load data, encompassing both regular and holiday-specific load profiles. Experimental results demonstrate that the MLR-GPR-GLSAR configuration achieves superior performance, with a Mean Absolute Percentage Error (MAPE) of 0.72% and an R^2 of 0.99, outperforming Generalized Additive Models (GAM), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LGB), Convolutional Long Short-Term Memory (CNN-LSTM), and Neural Hierarchical Interpolation for Time Series (NHITS), as baselines. The proposed method offers a scalable, interpretable solution suited to data-sparse operational environments. It highlights the value of correcting residual autocorrelation and using calendar-aware segmentation in improving load-forecasting accuracy.

INDEX TERMS Classification and regression trees (CART), day-ahead load forecasting (DALF), error correlation, explainable artificial intelligence (XAI), extreme gradient boosting (XGBoost), generalized additive models (GAM), Gaussian process regression (GPR), light gradient boosting machine (LGB), multiple linear regression (MLR), short term load forecasting (STLF).

NOMENCLATURE

ARIMA Autoregressive Integrated Moving Average.
CART Classification and Regression Tree.
CNN-LSTM Convolutional Long Short-Term Memory.

DoW Day of Week.
Bri-Hol Bridging-Holidays.
MLR Multiple Linear Regression.
LGB Light Gradient Boosting Machine.
GPR Gaussian Process Regression.
GLSAR Generalized Least Squares with Autoregressive Errors.
GAM Generalized Additive Models.

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LSTM	Long Short-Term Memory.
ML	Machine Learning.
NHITS	Neural Hierarchical Interpolation for Time Series.
SHAP	Shapely Additive exPlanations.
STLF	Short-Term Load Forecasting.
XAI	Explainable Artificial Intelligence.
XGBoost	Extreme Gradient Boosting.

I. INTRODUCTION

Accurate short-term load forecasting (STLF) remains a fundamental component in power system operation, influencing real-time dispatch, reserve scheduling, and grid reliability. However, the increasing penetration of distributed renewable generation, electrification of transport, and evolving consumption behavior have amplified temporal variability in electricity demand, particularly on holidays, weekends, and transitional “bridging” days. These fluctuations challenge the performance and interpretability of both conventional and deep-learning-based forecasting models.

Most existing STLF approaches rely on meteorological variables such as temperature, humidity, and solar irradiance to model nonlinear dynamics in load patterns [1], [2], [3], [4], [5]. While such models, including long short-term memory (LSTM)-based hybrids [2], ResNet-LSTM architectures [3], and boosted ensembles [6], [7], [8], [9], offer high accuracy, their dependency on external weather data poses several limitations: reduced scalability in data-scarce regions, high preprocessing complexity, and limited adaptability to operational constraints. Moreover, recent studies indicate that exogenous-feature-heavy models often lack interpretability, which is critical for power system operators [18], [19], [20].

Recent literature has proposed hybrid frameworks that incorporate calendar-based segmentation [10], [11], [12], [13], [14], seasonal decomposition [17], and feature-adaptive modeling [15] to overcome these limitations. However, many methods still overlook inter-day autocorrelation of period-specific residuals and tend to generalize holidays uniformly, missing variability across holiday types or pre-/post-holiday behaviors [11], [12], [13], [22]. Furthermore, while Generalized Least Squares with Autoregressive Errors (GLSAR) [21] and Explainable AI (XAI) techniques, such as SHAP (Shapely Additive exPlanations) [19], offer promising directions for post-hoc analysis, they are rarely integrated into end-to-end operational pipelines.

In parallel, there has been a recent surge of interest in explainable and probabilistic forecasting. Reference [34] provides a comprehensive review showing how interpretability and uncertainty quantification are becoming central to modern load forecasting. User-centric probabilistic frameworks, such as those introduced by Robin et al. [35], incorporate explainable gradient boosting models to improve transparency for operators. Similarly, Muntasir et al. [36] propose an explainable multivariate LSTM for long-term forecasting that couples probabilistic outputs with interpretable drivers.

Attention-based and federated approaches have also been combined with explainability; for example, [37] Sarker et al. (2024) integrated SHAP-based interpretability into secure smart grid forecasting pipelines. Most recently, [38] demonstrated interpretable probabilistic prediction in decentralized smart grids, underscoring the potential of explainable AI to build trust in probability-informed decision making. Collectively, these studies highlight the need to extend deterministic explainable forecasting toward probabilistic frameworks, which we identify as a promising direction for future research.

To address the limitations of existing STLF models, such as their dependence on weather data, limited interpretability, and inadequate handling of calendar effects, this study proposes a novel calendar-aware, explainable, and weather-independent STLF framework. First, a Classification and Regression Tree (CART) is used to segment load behavior by day of week, holiday, and bridging holiday indicators, with mean squared error (MSE) as the splitting criterion to capture high-variance calendar patterns. For each segment, separate forecasting models are trained using only lagged load features (previous day, previous week, and two weeks prior). These models include Multiple Linear Regression (MLR), XGBoost, LightGBM (LGB), Convolutional Long Short-term Memory (CNN-LSTM), Neural Hierarchical Interpolation for Time Series (NHITS), and Generalized Additive Models (GAM). Gaussian Process Regression (GPR) is selectively integrated for holiday-related nodes that exhibit strong nonlinearity, thereby improving accuracy and uncertainty quantification. Residuals from all models are further refined using GLSAR to account for inter-day autocorrelation across 48-period intervals. This refinement step helps mitigate error propagation and supports consistent performance across daily time steps. To ensure interpretability, the model integrates SHAP, which quantifies the marginal contribution of each input feature to the forecasting performance of calendar-based load groups. This allows the evaluation of feature influence across calendar segments and time periods, promoting a better understanding of how input variables affect forecasts without requiring additional external data [19], [20].

Thus, the key contributions of this study are:

- A weather-independent load forecasting method that maintains accuracy using only historical and calendar data.
- GLSAR captures inter-day autocorrelated residuals, improving final forecast fidelity.
- A hybrid GPR-enhanced strategy that boosts performance in holiday load groups with high nonlinearity.
- A complete interpretability framework using SHAP to trace variable influence across load categories.

The rest of this paper is structured as follows. Section II describes the methodology, including data preprocessing, classification, and forecasting procedures. Section III presents the experimental results and discusses their implications. Finally, Section IV offers concluding remarks.

II. METHODOLOGY

This study proposes an explainable, calendar-driven DALF framework comprising four modules as in Fig. 3: (1) CART segmentation, (2) segment-specific forecasting, (3) GLSAR residual correction, and (4) SHAP-based interpretability. Model hyperparameters are optimized via grid search, and performance is evaluated using walk-forward validation with an expanding window to ensure robustness.

A. DATA COLLECTION AND PREPROCESSING

The primary dataset comprises half-hourly electricity demand data from the Electricity Generating Authority of Thailand (EGAT), covering the period from January 2019 to December 2021. Each day has 48 periods, yielding 52,608 observations. After parsing dates, we normalize load values and encode three calendar variables: day of the week, holiday flag, and bridging holiday flag. These features capture recurring demand shifts associated with weekly cycles and special calendar events.

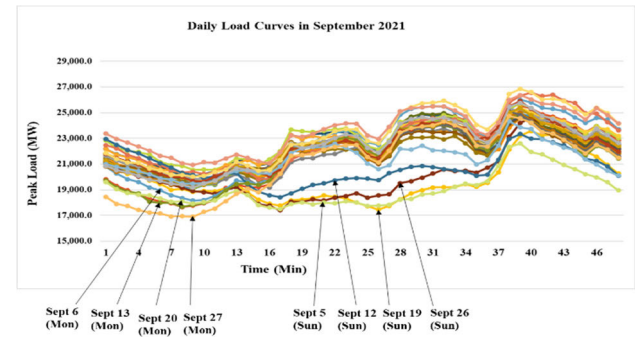


FIGURE 1. Daily load curves (48 Periods) in September 2021.

As shown in Fig. 1 for September 2021, Thailand’s daily load patterns exhibit consistent peaks in the late afternoon and early evening, followed by a decline overnight. This typical electricity usage pattern reflects a higher demand for lighting, cooling, and industrial processes during waking hours. These fluctuations, influenced by the difference between weekdays and weekends, underscore the complexity of electricity consumption. Models must accurately reflect these variations. Introducing calendar-driven features, such as holidays and seasonal variations, is crucial for capturing these intricate patterns.

To further validate model generalization, the proposed CART-MLR-GPR-GLSAR framework was applied to France’s national load dataset (2021–2023, 24 periods/day). This dataset exhibits characteristics different from those of Thailand, including sharper winter peaks and more moderate summer loads, due to its temperate climate.

In France, the electricity load profile follows a distinct weekday–weekend pattern typical of temperate European systems. Fig. 2 illustrates daily electricity load curves for the first week of June 2021. Weekdays (Monday–Friday) share similar profiles—demand rises sharply from 6:00 to noon,

with a secondary evening peak around 19:00–21:00 due to residential activity.

Weekends, especially Sunday, exhibit significantly lower, smoother demand because the commercial and industrial sectors remain closed. Seasonally, winter loads are elevated by heating; summer is moderate; spring/autumn are balanced. These dynamics highlight how climate and lifestyle jointly shape France’s electricity use. Identical preprocessing, CART segmentation, and GLSAR procedures were applied to ensure methodological consistency.

Both datasets were processed under the same segmentation and walk-forward protocol to ensure methodological consistency. This cross-regional validation supports assessing model transferability across grid structures, seasonal regimes, and temporal resolutions.

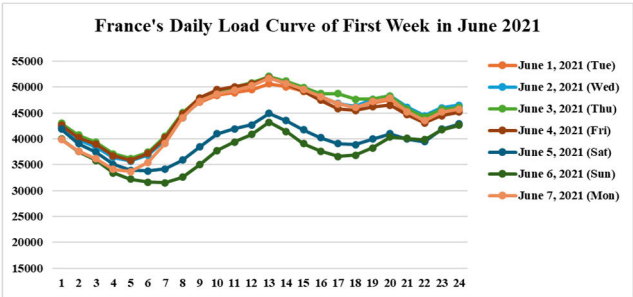


FIGURE 2. France’s daily load curves of the first week in June 2021.

B. CALENDAR-BASED CART SEGMENTATION

A multi-output regression CART groups days by similarity in their daily load profiles, using three calendar indicators—day of week, holiday, and bridging holiday—as predictors. Table 1 presents the arrangement of sample data used for load data classification, including the independent and target variables.

TABLE 1. Illustrative structure of input and target variables for load data classification.

No:	Date	Independent Variables			Target Variable (Peak Load)			
		DoW	Hol	Bri_Hol	L_{t1}	L_{t2}	-----	L_{t48}
1	4/15/2019	1	1	0	19864.1	19352.3	-----	20597.9
2	4/16/2019	2	1	0	20067.5	19516.5	-----	22515.0
3	4/17/2019	3	0	0	21651.7	21390.6	-----	25056.7
4	4/18/2019	4	0	0	24219	23687.2	-----	26402.4
5	4/19/2019	5	0	0	25587.1	25262.7	-----	27304.5
6	4/20/2019	6	0	0	26334.7	25955.8	-----	27498.4
7	4/21/2019	7	0	0	26684.8	26019.2	-----	25817.4
8	4/22/2019	1	0	0	25157.5	24497.8	-----	25788.1
9	4/23/2019	2	0	0	24614.8	23808.9	-----	27238.0

Training on the 2019–2020 dataset yields leaf nodes that isolate homogeneous segments (regular Mondays, holiday Mondays, bridging holiday Fridays). Once the tree is grown,

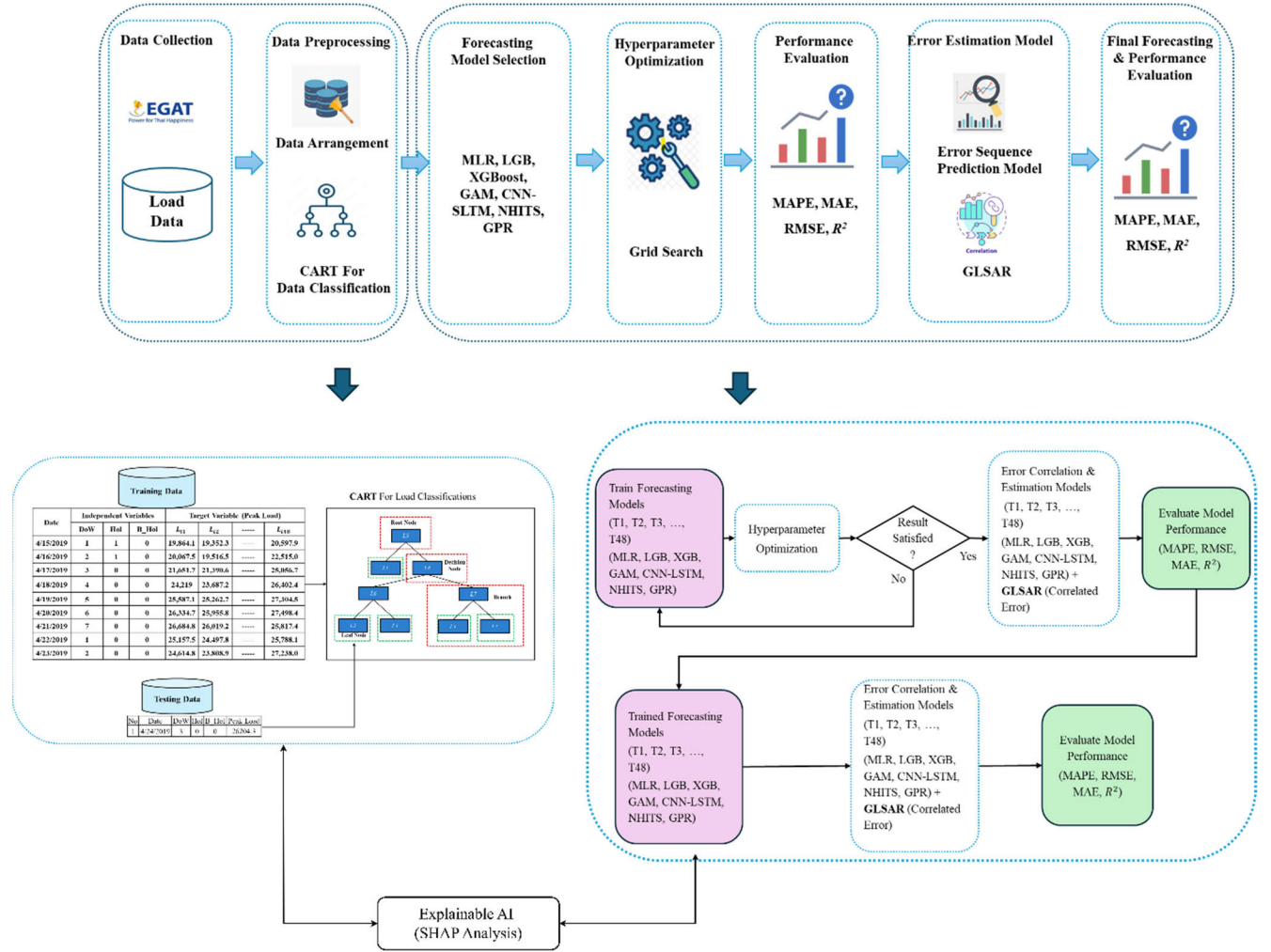


FIGURE 3. Proposed framework architecture for calendar-aware load forecasting with integrated explainability.

each 2021 observation is routed to its corresponding leaf for proper out-of-sample grouping [10]. This calendar-aware clustering captures systematic load variations and prepares tailored subgroups for downstream forecasting models [26].

At each internal node n , CART considers splits on Day-of-Week (DoW), Holiday (Hol), or Bridging-Holiday (Bri_Hol) by minimizing the weighted sum of intra-node mean squared error over all 48 half-hour periods. Specifically, for node n containing s_n days, the MSE is written as in (1):

$$MSE = \frac{1}{s_n} \sum_{d \in n} \sum_{t=1}^{48} \left(L_t^d - \bar{L}_t^n \right)^2 \quad (1)$$

where s_n is the number of days in node n , L_t^d is the actual load on day d at period t , and $\bar{L}_t^n$ is the mean load at period t across all days in node n .

For any candidate who splits the dividing node n into left and right children, the impurity measure is as in (2):

$$MSE_{split} = \frac{s_{left}}{s_n} MSE_{left} + \frac{s_{right}}{s_n} MSE_{right} \quad (2)$$

where s_{left} and s_{right} are the numbers of days in the left and right child nodes, respectively. The split yielding the smallest MSE_{split} is chosen, thereby grouping days with highly similar intra-day load profiles.

C. DATA SEGMENTATION AND FORECAST AGGREGATION

Within each leaf, the forecasting models use three lagged inputs: load from one day before $L_t^s(d-1)$, one week before $L_t^s(d-7)$, and two weeks before $L_t^s(d-14)$ to forecast each half-hour period via forecasting models. These lagged values are chosen to reflect short-term and weekly consumption trends. Some calendar segments contain sparse samples (national holidays, religious festivals, bridging days). For these nodes, specifically Leaf Nodes 2, 7, 8, 12, 13, 14, and 15, the dataset contains insufficient training size for linear models. Gaussian Process Regression (GPR) replaces MLR for leaf nodes with sparse holiday data to better capture non-linearities. This study also benchmarks LightGBM (LGB), XGBoost (XGB), and Generalized Additive Models (GAM) under identical hyperparameter ranges.

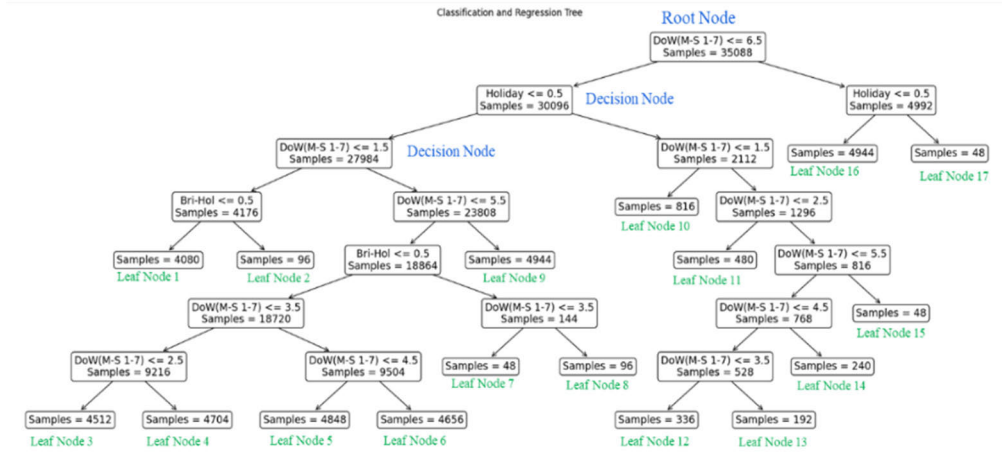


FIGURE 4. Structure of Classification and Regression Tree (CART) for training data segmentation.

Table 2 presents an illustrative training–testing split for predicting January 8, 2021, from leaf node 6, for each half-hour period t . Although Table 2 shows a “10-day” window for clarity, in our full implementation, each leaf-node model is trained on the complete two-year (2019–2020) history before forecasting 2021. This rolling-window, walk-forward approach ensures that every test day benefits from all past observations available within its calendar-defined segment, as shown in Fig. 5.

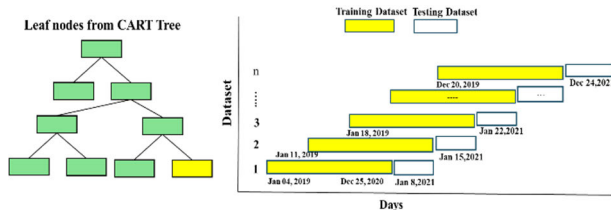


FIGURE 5. Walk-forward validation routine for robust daily load forecasting.

This testing framework ensures that the forecasting models are continuously updated with the most recent and relevant data, simulating real-world operational forecasting scenarios. To generate the final daily load forecast, each of the 48 half-hour periods of the test day is predicted individually using the corresponding trained model. Once the 48-point forecasts are obtained, the average of these 48 predicted values is computed to represent the aggregated daily load forecast. This approach preserves temporal resolution and enhances the reliability of the overall forecast by capturing intra-day load dynamics.

D. FORECASTING MODEL DEVELOPMENT

Following the CART segmentation, five forecasting approaches were developed and evaluated for each leaf-node group g and a half-hour period t . Each model uses three

TABLE 2. Training and testing data structure for cart leaf node-based forecasting for January 8, 2021.

No.	Inputs			Target
	$L_t^g(d-14)$	$L_t^g(d-7)$	$L_t^g(d-1)$	$F_t^g(d)$
Training Dataset	09/25/2020 (Fri)	10/02/2020 (Fri)	10/08/2020 (Thu)	10/09/2020 (Fri)
	10/02/2020 (Fri)	10/09/2020 (Fri)	10/15/2020 (Thu)	10/16/2020 (Fri)
	...	...	...	...
	...	...	...	...
	...	...	...	...
	...	...	...	...
	...	...	...	...
	...	...	...	...
	...	...	...	...
	12/11/2020 (Fri)	12/18/2020 (Fri)	12/24/2020 (Thu)	12/25/2020 (Fri)
Testing Dataset	12/18/2020 (Fri)	12/25/2020 (Fri)	01/07/2021 (Thu)	01/08/2021 (Fri)
	...	...	...	...

lagged inputs: load from days $(d-1)$, $(d-7)$, and $(d-14)$ to predict the next day’s load $F_t^g(d)$. Model parameters were optimized through grid search over predefined hyperparameter ranges and assessed using walk-forward validation. The forecasting equation for all experiments is given in (3).

$$F_t^g(d) = a_{0,t}^g + a_{1,t}^g L_t^g(d-14) + a_{2,t}^g L_t^g(d-7) + a_{3,t}^g L_t^g(d-1) \quad (3)$$

where $L_t^g(d)$ is the electricity load (MW) in the leaf node group g at time t on day d , $F_t^g(d)$ is a forecasted load, and $\{a_{i,t}^g\}$ are model coefficients. The use of lagged features from 1-, 7-, and 14-days capture both short-term dependencies and weekly periodicity.

These models learn complex relationships between past observations and future demand, improving accuracy over purely linear baselines.

1) MULTIPLE LINEAR REGRESSION (MLR)

The MLR model serves as a transparent statistical baseline, estimating coefficients in (3) via ordinary least squares. It assumes linear dependence between lagged load values

and the forecasted value, without additional hyperparameters or regularization. This interpretability makes MLR a stable reference point for comparison against more flexible models.

2) GENERALIZED ADDITIVE MODEL (GAM)

As shown in (4), the GAM extends the linear framework by replacing each lagged term with a nonparametric smooth function, allowing flexible nonlinear relationships:

$$F_t^g(d) = \beta_{0,t}^g + \sum_{j=1}^3 f_{j,t}^g(L_t^g(d - l_j) + \varepsilon_t^g(d) \quad (4)$$

where $\varepsilon_t^g(d) \sim \mathcal{N}(0, \sigma^2)$ and each smooth function $f_{j,t}^g(\cdot)$ is estimated via penalized iteratively reweighted least squares, enabling the model to capture complex nonlinear relationships between past load values and future demand.

3) GRADIENT BOOSTING MODELS (LGB, XGBoost)

For both LGB and XGBoost, input features are standardized using z-score normalization (fitted on the training data and applied to the test data) to improve convergence. Both models were tuned via a grid search over key hyperparameters, including the number of trees $\in \{100, 200, 300\}$, learning rate $\in \{0.05, 0.1, 0.5\}$, maximum tree depth $\in \{3, 5, 7\}$, subsampling ratios $\in \{0.7, 0.8, 0.9\}$, and `colsample_bytree` $\in \{0.8, 0.9, 1.0\}$ to identify the configuration yielding the lowest cross-validated MAPE. The selected settings (100 trees, learning rate = 0.1, max depth = 5, subsample = 0.8, `colsample_bytree` = 0.8) emerged from this search. Early stopping (10 rounds) based on validation MAPE prevented over-fitting.

By using the same hyperparameters for both models, the models ensure that performance differences reflect algorithmic design rather than tuning discrepancies.

4) GAUSSIAN PROCESS REGRESSION (GPR)

To handle sparse holiday segments, the GPR model is used as a non-parametric model with a kernel consisting of a constant term multiplied by a Matern covariance (length scale = 20, $\nu = 1.5$) plus a white-noise component. The model is optimized with ten restarts of the marginal-likelihood optimizer. When a leaf node's sample size falls below a predefined threshold, we borrow kernel hyperparameters from the most similar segment or interpolate historical forecasts to maintain stability.

5) CONVOLUTIONAL NEURAL NETWORK–LSTM (CNN–LSTM)

The CNN–LSTM model was developed to integrate the complementary strengths of convolutional and recurrent architectures in short-term load forecasting. Convolutional layers serve as local feature extractors, identifying short-term fluctuations and repetitive motifs in the daily load sequence. The CNN block is configured with 32 filters (kernel size 5, dropout 0.2), and the LSTM block comprises two layers with 64 hidden units each (dropout 0.2).

The CNN–LSTM integrates convolutional and recurrent layers to extract short-term features and long-range dependencies from 84-day sliding input windows using the same three lagged load inputs ($(d - 1)$, $(d - 7)$, and $(d - 14)$). The model was trained with Adam (learning rate = 0.003) and early stopping on validation loss to ensure stability and prevent overfitting.

6) NEURAL HIERARCHICAL INTERPOLATION FOR TIME SERIES (NHITS)

The NHITS model extends N-BEATS by introducing hierarchical interpolation to represent multi-scale temporal patterns in electricity load. Each block captures coarse-to-fine components of demand, enabling accurate modeling of long- and short-term variations.

Using three lagged load inputs, the NHITS was tuned through a grid search over input size $\in \{28, 56, 84\}$, n blocks $\in \{1, 2, n\}$, n layers $\in \{1, 2\}$, dropout $\in \{0.0-0.2\}$, and learning rate $\in \{5 \times 10^{-4} - 2 \times 10^{-3}\}$. Models were trained with Adam for ≤ 80 epochs using walk-forward validation and early stopping.

E. ERROR-CORRELATION CORRECTION WITH GLSAR

To address residual autocorrelation in the base forecasting models, a Generalized Least Squares with Autoregressive Residuals (GLSAR) correction is applied. This procedure models the residual sequence for each period as an autoregressive process across consecutive days, capturing interday error dependencies. For each leaf node group g and period t , the residual on day d is given by:

$$r_t^g(d) = L_t^g(d) - F_t^g(d) \quad (5)$$

where $L_t^g(d)$ and $F_t^g(d)$ denote the actual and forecasted loads, respectively. Using the most recent seven residuals for the same period t and group g , a GLSAR model is dynamically fitted to estimate an autoregressive structure (typically AR (1)) as follows:

$$r_{pred_t}^g(d) = a_{4,t}^g + b_{1,t} r_t^g(d - 1) \quad (6)$$

where, $a_{4,t}^g$ and $b_{1,t}$ are GLSAR parameters estimated via generalized least squares. The term $r_t^g(d - 1)$ represents the previous-day residual within the same calendar-defined group.

The corrected forecast is obtained by incorporating the predicted residual into the base model:

$$F_{GLSAR_t}^g(d) = a_{0,t}^g + a_{1,t}^g L_t^g(d - 14) + a_{2,t}^g L_t^g(d - 7) + a_{3,t}^g L_t^g(d - 1) + r_{pred_t}^g(d) \quad (7)$$

This residual correction loop refines the base forecasts (MLR, LGB, XGB, GAM, NHITS, CNNLSTM, and GPR) by compensating for interday error persistence, thus improving stability and accuracy across calendar-based load segments.

F. INTERPRETING MODEL PREDICTIONS VIA SHAP ANALYSIS

To enhance the explainability of our short-term load-forecasting framework, we employed SHAP, a widely used XAI technique for quantifying feature influence. Since the calendar-based features — Day of Week, Holiday, and Bridging Holiday — are not directly used as inputs to the regression models but instead guide the CART-based data segmentation, a surrogate modeling approach was adopted. Specifically, a surrogate XGBoost regressor was trained using the CART splitting variables as input and the MLR-predicted load values as output. This captures the implicit mapping from calendar-based segmentation to forecasted load. The SHAP values, which quantify the marginal contribution of each input variable to a model prediction, were computed as follows:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (8)$$

where F is the complete set of features, and S is a subset excluding i . This approach ensures local accuracy, consistency, and additivity, thus offering an interpretable mechanism to explain how temporal segmentation influences downstream forecasts.

III. RESULT AND DISCUSSION

A. EVALUATION METRICS

Forecasting performance was evaluated using four standard metrics: Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). These indicators, computed over the 48 half-hourly intervals per day, quantify both absolute and relative deviations between observed and predicted loads, as defined in (9) – (12).

Higher R^2 and lower MAPE, MAE, and RMSE values indicate stronger forecast fidelity and reliability across temporal segments.

$$MAPE = \frac{1}{48} \sum_{t=1}^{48} \left| \frac{L_t^g(d) - F_t^g(d)}{L_t^g(d)} \right| \times 100 \quad (9)$$

$$MAE = \frac{1}{48} \sum_{t=1}^{48} |L_t^g(d) - F_t^g(d)| \quad (10)$$

$$RMSE = \sqrt{\frac{1}{48} \sum_{t=1}^{48} (L_t^g(d) - F_t^g(d))^2} \quad (11)$$

$$R^2 = 1 - \frac{\sum_{t=1}^{48} (L_t^g(d) - F_t^g(d))^2}{\sum_{t=1}^{48} (L_t^g(d) - \overline{L_t^g(d)})^2} \quad (12)$$

where $\overline{L_t^g(d)}$ is the mean actual load.

B. IMPACT OF GLSAR ON FORECASTING PERFORMANCE

Table 3 summarizes the effect of the Generalized Least Squares with Autoregressive Residuals (GLSAR) correction on different model configurations. Across all architectures, GLSAR consistently reduced forecasting errors by 55 – 80%

and improved explanatory power by 0.15 – 0.45 R^2 units on average. For example, the MLR–GPR hybrid reduced MAPE from 3.24% to 0.72% and increased R^2 from 0.81 to 0.99.

TABLE 3. Performance comparison of forecasting models classified by cart with and without GLSAR error correction.

Model / Metric	GLSAR Applied	MAPE (%)	RMSE (MW)	MAE (MW)	R^2
MLR-GPR	No	3.24	991.97	689.07	0.81
	Yes	0.72	232.02	154.12	0.99
XGBoost-GPR	No	3.76	1063.43	798.15	0.77
	Yes	1.21	399.63	259.90	0.97
LGB-GPR	No	3.73	1060.15	790.05	0.77
	Yes	1.21	396.37	259.20	0.97
GAM-GPR	No	3.78	1080.13	798.97	0.77
	Yes	0.90	322.51	191.67	0.98
CNN-LSTM - GPR	No	5.15	1,368.86	1,126.44	0.55
	Yes	3.54	972.58	769.23	0.76
NHITS-GPR	No	6.17	1,480.20	1,305.57	0.51
	Yes	4.09	1,098.53	883.888	0.66

Similarly, ensemble models such as XGBoost–GPR and LGB–GPR improved their R^2 values, exceeding 0.97 after GLSAR correction. Even deep learning methods, including CNN–LSTM–GPR and NHITS–GPR, exhibited notable reductions in error, highlighting GLSAR’s ability to mitigate temporal autocorrelation in residuals. Overall, integrating GLSAR significantly enhances stability and robustness across statistical, ensemble, and neural architectures by addressing inter-day residual dependencies.

C. MONTHLY FORECASTING STABILITY

The monthly performance comparison (Fig. 6 and Table 4) demonstrates the proposed framework’s robustness under seasonal variability. Without error correction, MLR models exhibited performance fluctuations between 1.79% (June) to 4.32% (January) MAPE depending on the month.

After applying GLSAR, the monthly MAPE remained below 0.75%, while the R^2 exceeded 0.97 across all months.

GAM–GLSAR achieved the lowest average MAPE (0.83%) and RMSE (287.8 MW), confirming that combining smooth nonlinear modeling with residual correction provides stable, year-round performance. These results validate the model’s ability to maintain accuracy despite variations in consumption behavior between summer and winter months.

D. FORECASTING ON HOLIDAY AND BRIDGING-HOLIDAY LOAD GROUPS

Forecasting during holidays and bridging periods presents unique challenges due to high variability and limited historical data. As shown in Table 5, the GPR-only model achieved an average MAPE of 5.44% and R^2 of 0.64.

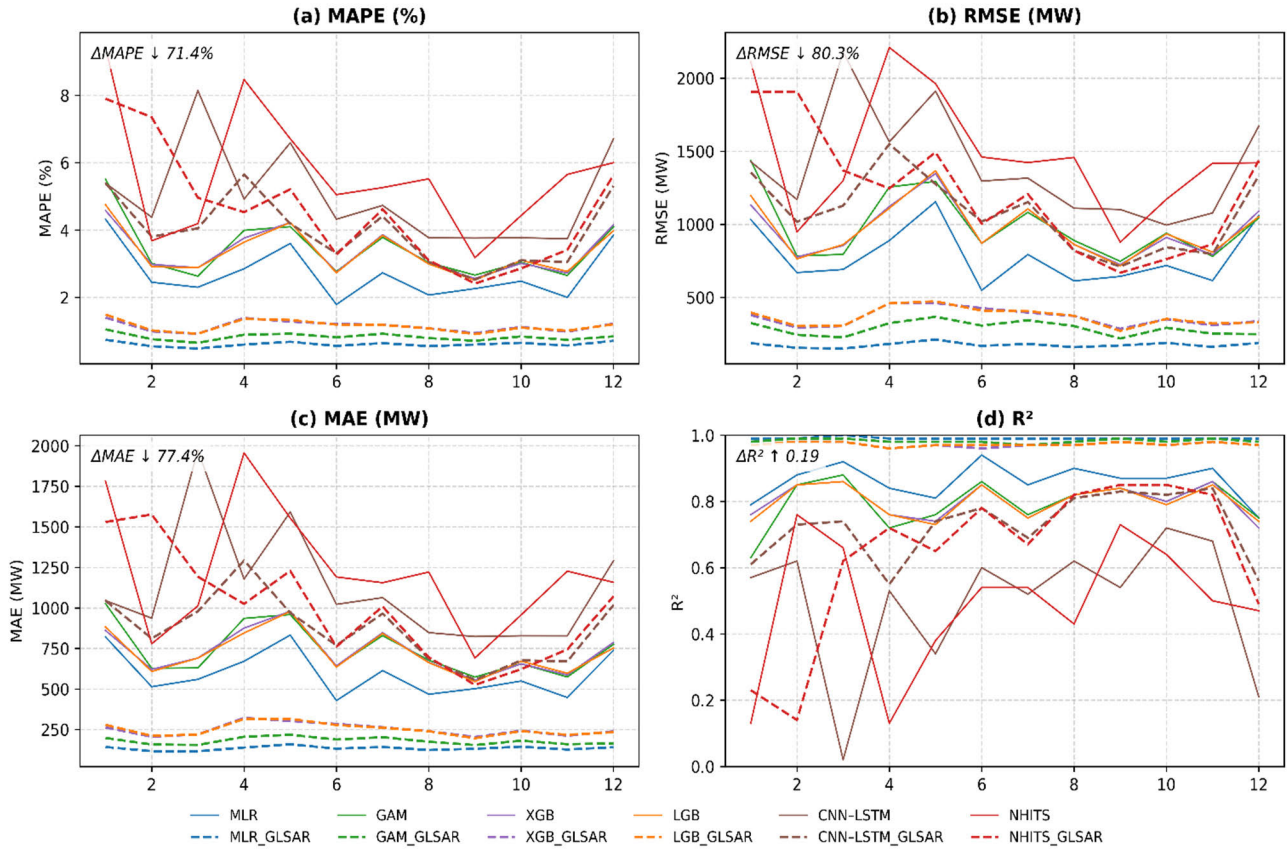


FIGURE 6. Monthly forecasting performance comparison before and after GLSAR for normal-day load groups.

TABLE 4. Monthly forecasting performance of proposed model before and after GLSAR for normal day load groups classified by CART.

Month	MLR				MLR - GLSAR			
	MAPE (%)	RMSE (MW)	MAE (MW)	R ²	MAPE (%)	RMSE (MW)	MAE (MW)	R ²
1	4.32	1032.89	820.92	0.79	0.74	186.87	142.50	0.99
2	2.45	669.76	513.82	0.88	0.55	154.98	116.43	0.99
3	2.30	691.46	559.46	0.92	0.48	150.156	115.63	1.00
4	2.85	889.02	670.83	0.84	0.60	182.097	139.18	0.99
5	3.60	1154.61	832.20	0.81	0.68	211.68	158.94	0.99
6	1.79	548.61	428.42	0.94	0.56	168.23	132.07	0.99
7	2.73	793.62	613.34	0.85	0.64	182.44	142.23	0.99
8	2.07	613.02	467.50	0.90	0.55	159.68	123.44	0.99
9	2.26	643.99	501.86	0.87	0.60	171.09	131.36	0.99
10	2.48	718.43	548.87	0.87	0.65	188.73	144.25	0.99
11	2.00	615.08	447.36	0.90	0.57	161.79	126.97	0.99
12	3.85	1059.57	738.83	0.75	0.71	189.13	140.85	0.99
Avg/leaf	2.72	785.84	595.28	0.86	0.61	175.57	134.49	0.99

After integrating GLSAR, accuracy improved dramatically to 1.27% MAPE and $R^2 = 0.96$, reflecting over 75% reduction in relative error. The most pronounced improvements occurred during January and April, coinciding with major national holidays. These findings confirm that the GPR–GLSAR cascade effectively stabilizes predictions in data-sparse, high-variance segments, thereby offering reliable forecasting for special-event days.

Forecasting electricity load during holidays presents unique challenges due to high variability and limited historical data. Table 5 summarizes the model performance for holidays and bridging holiday segments. GPR alone yielded an average MAPE of 5.44% and an R^2 of 0.64, reflecting its difficulty in capturing the high variance inherent in these periods. However, when combined with GLSAR, the performance improved substantially: MAPE decreased to 1.27%,

TABLE 5. Forecasting performance on holiday & Bri-holiday load group classified by CART.

Month	GPR				GPR - GLSAR			
	MAPE	RMSE	MAE	R ²	MAPE	RMSE	MAE	R ²
1	7.51	1,118.89	940.06	0.65	2.10	468.70	271.76	0.94
2	2.95	797.24	614.22	0.86	0.86	237.55	177.07	0.99
3								
4	9.67	1,724.02	1,794.45	0.37	1.44	427.31	284.49	0.98
5	7.55	1,407.56	1,745.17	0.34	1.61	624.95	373.70	0.90
6	4.88	1,400.16	1,239.37	0.67	1.14	398.25	292.62	0.97
7	4.78	1,058.13	979.23	0.68	1.45	358.51	285.40	0.95
8	4.93	1,161.98	1,050.38	0.62	0.91	278.93	196.66	0.98
9	4.10	942.12	912.71	0.68	0.90	426.50	197.19	0.94
10	3.16	936.38	701.25	0.67	0.83	281.73	187.28	0.97
11								
12	4.83	1,033.02	781.51	0.87	1.48	324.69	247.04	0.95
Avg/leaf	5.44	1,237.47	1,075.84	0.64	1.27	380.09	251.32	0.96

and R², increased to 0.96. The most pronounced improvements occurred during January and April, which include high-load anomalies due to national holidays. These results emphasize the utility of the GPR-GLSAR cascade in data-sparse, high-variance conditions.

E. CROSS-DATASET VALIDATION

To assess generalization, the framework was validated using an independent dataset from France (2021–2023) with 24 intervals per day. Table 6 shows that GLSAR improved accuracy across nearly all models. The MLR–GPR–GLSAR model achieved a 25% reduction in MAPE (3.86% to 2.90%) and increased R² from 0.82 to 0.96. France exhibits sharper winter peaks, stronger weekday–weekend contrast, and more stable holiday behavior compared to Thailand.

TABLE 6. Performance comparison of forecasting models classified by CART with and without GLSAR error correction.

Model / Metric	GLSAR Applied	MAPE (%)	RMSE (MW)	MAE (MW)	R ²
MLR-GPR	No	3.86	2,771.32	2,188.31	0.82
	Yes	2.90	2,091.72	1,527.02	0.96
XGBoost-GPR	No	4.95	3,553.09	2,548.84	0.79
	Yes	3.24	2,257.69	1,449.93	0.84
LGB-GPR	No	8.30	5,180.57	4,157.02	0.772
	Yes	8.26	5,157.92	4,143.81	0.774
GAM-GPR	No	3.76	2,767.39	2,145.73	0.82
	Yes	3.17	2,612.09	1,776.10	0.94
CNN-LSTM - GPR	No	7.28	4,944.37	3,877.35	0.81
	Yes	5.01	3,856.13	2,758.56	0.88
NHITS-GPR	No	25.32	13,654.27	11,841.05	-0.5
	Yes	6.27	4,940.51	3,480.22	0.79

Despite these differences, GLSAR consistently improved model accuracy. For example, MLR–GPR improved from 3.86% to 2.90% MAPE (R² from 0.82 to 0.96). CNN–LSTM–GPR improved from 7.28% to 5.01%, while NHITS–GPR improved substantially from 25.32% to 6.27%. Error distribution analysis showed that GLSAR reduced the spread between the 50th and 95th percentiles by 40–70%, indicating reduced extreme deviations.

F. ERROR CORRELATION ANALYSIS AND STATISTICAL VALIDATION

Electricity-load residuals often exhibit serial dependence across consecutive intra-day periods (48 per day in Thailand). Such dependence violates the independence assumption underlying ordinary regression and kernel estimators, potentially biasing parameter estimates and underestimating error magnitudes. We therefore quantify residual autocorrelation before and after GLSAR correction using two complementary indicators: the Durbin–Watson (DW) statistic and the lag-1 autocorrelation (ACF1). Across CART leaf nodes, residuals before GLSAR show positive dependence (DW median $\approx$ 1.79, ACF1 median $\approx$ -0.02 with several nodes positive), confirming serial correlation at intra-day timescales. After GLSAR, DW values shift toward the ideal value of 2, and ACF1 values move closer to zero, indicating effective whitening of residuals.

Table 7 reports node-level DW/ACF1 summaries (Thailand). Median improvements of Δ DW = +0.03 and Δ ACF1 = +0.04 demonstrate measurable error-correlation reduction even where baseline autocorrelation is moderate—a known characteristic of Thailand’s climate-sensitive but relatively weather-independent demand. To confirm that these corrections translate into accuracy gains, we conducted paired statistical tests comparing pre- and post-GLSAR errors, aggregated across all nodes. Table 8 shows that improvements are statistically significant for MAPE, MAE, RMSE, and R² under both paired t-tests and Wilcoxon signed-rank tests ($p < 0.01$ in all cases). These results

TABLE 7. Error correlation analysis using durbin–watson and ACF1 before vs. After GLSAR for normal leaves.

leaf	series	DW_before	DW_after	DW_delta	ACF1_before	ACF1_after	ACF1_delta
1	48	1.75	1.68	0.03	-0.06	0.07	0.11
3	48	1.52	1.59	0.06	0.11	0.13	0.04
4	48	1.74	1.79	0.05	-0.03	0.03	0.05
5	48	1.66	1.75	0.07	0.02	0.0003	-0.01
6	48	1.89	1.94	0.02	-0.01	0.01	0.04
9	48	2.02	1.93	-0.14	-0.12	-0.01	0.12
10	48	0.07	1.84	1.77	0.96	0.07	-0.9
11	48	0.08	1.83	1.75	0.96	0.07	-0.9
16	48	1.86	1.83	0.04	0.02	0.03	-0.001
ALL		1.79	1.81	0.03	-0.02	0.03	0.04

TABLE 8. Error correlation analysis using durbin–watson and ACF1 Before vs. After GLSAR for sparse data leaves.

leaf	series	DW_before	DW_after	DW_delta	ACF1_before	ACF1_after	ACF1_delta
2	48	0.07	0.67	0.60	0.97	0.59	-0.37
7	48	0.05	0.96	0.91	0.98	0.47	-0.50
8	48	0.11	0.60	0.48	0.94	0.65	-0.29
12	48	0.09	0.84	0.75	0.94	0.58	-0.36
13	48	0.05	0.80	0.75	0.96	0.60	-0.37
14	48	0.11	0.78	0.67	0.94	0.60	-0.34
15	48	0.05	0.77	0.72	0.95	0.60	-0.35
ALL		0.07	0.67	0.60	0.97	0.59	-0.37

TABLE 9. Statistical significance of post-GLSAR improvements (Thailand).

Metric	n	Test	Statistic	p-value (one-sided)	Significance ($\alpha = 0.05$)	Interpretation
MAPE (%)	16	Paired <i>t</i> -test	$t = 6.842$	< 0.0001	✓	GLSAR significantly reduces MAPE
		Wilcoxon signed rank	$W = 0.00$	< 0.0001	✓	Confirms robust improvement
MAE	16	Wilcoxon signed rank	$W = 136$	< 0.0001	✓	Error magnitude reduction is significant.
RMSE	16	Wilcoxon signed rank	$W = 136$	< 0.0001	✓	Forecast dispersion has reduced significantly.
R ²	16	Paired <i>t</i> -test	$t = -4.92$	0.0002	✓	GLSAR significantly increases R ²

provide quantitative evidence that GLSAR materially improves residual independence and forecasting reliability.

To complement the analysis for normal load nodes, Table 8 presents DW and ACF1 statistics for holiday-related leaf nodes where GPR is used. These nodes exhibit more substantial residual autocorrelation due to sparse training samples and irregular holiday behavior. GLSAR reduces ACF1 by 0.29–0.50 and increases DW by 0.48–0.91, confirming its essential role in stabilizing forecasts in low-data conditions.

Table 8 reports DW and ACF1 statistics for holiday and sparse-data CART leaf nodes. Before GLSAR, residual sequences are highly autocorrelated, with DW values of 0.04–0.11 and ACF1 values exceeding 0.94, indicating extremely strong serial dependence. After applying GLSAR, DW values increase substantially ($\Delta DW = 0.48$ –0.91), while ACF1 decreases by 0.29–0.50, demonstrating effective removal of autocorrelation even in highly sparse nodes. These corrections directly improve the stability of GPR-based forecasts used in holiday leaves, reducing MAPE by 65–93% across these nodes.

To assess whether these improvements translate into significant forecasting gains, we performed paired *t*-tests and Wilcoxon signed-rank tests comparing MAPE, MAE, RMSE, and R² before and after GLSAR across all 16 leaf nodes. As shown in Table 9, both tests yield $p < 0.001$, confirming that the accuracy improvement is statistically significant rather than incidental. To verify the statistical significance of the performance improvements, paired *t*-test and Wilcoxon signed-rank tests were applied across 12 months of results. Paired tests confirm that post-GLSAR performance improvements are statistically significant at $\alpha = 0.05$.

To evaluate whether error correlation is a dataset-specific phenomenon or a structural property of electricity load forecasting, we also analyzed residual autocorrelation on the France dataset. Like Thailand, the French residuals exhibited strong positive dependence prior to GLSAR, with DW values ranging from 1.10 to 1.35 and ACF1 values between 0.45 and 0.78. After GLSAR, DW increased toward the ideal value (1.95–2.24), and ACF1 decreased to 0.01–0.06. These results confirm that serial error correlation is present across

different countries, climatic conditions, and temporal resolutions (48 vs. 24 periods/day), and that GLSAR consistently removes this dependence. This validates the generalizability of the proposed framework beyond the Thailand dataset.

G. MODEL INTERPRETABILITY VIA SHAP ANALYSIS

To ensure transparency, SHAP—a leading Explainable AI technique—was used to interpret both calendar-based segmentation features and lagged load inputs. Since the segmentation variables (Day of Week, Holiday, Bridging Holiday) are not direct inputs to the forecasting models, a surrogate XGBoost regressor was trained to approximate the MLR outputs using these calendar variables. The surrogate achieved an R^2 of 0.983, confirming its fidelity.

Fig. 7 summarizes SHAP values across all segments, indicating that Day of Week (DoW) is the dominant factor influencing load forecasts. Holiday flags consistently contributed to demand reductions, while Bridging Holidays showed minor effects. This aligns with domain expectations: industrial and commercial activity dips during public holidays, whereas bridging holidays only partially affect demand.

Figs. 8 and 9 illustrate instance-level SHAP waterfall plots for a normal day and a Songkran holiday, respectively. On April 13, 2021 (Songkran), the Holiday variable led to a sharp negative adjustment (−1920 MW), while weekday activity (DoW = Tuesday) added +1165 MW. In contrast, a typical Monday showed smaller SHAP shifts tied to weekday effects. These visualizations demonstrate how SHAP isolates the impact of overlapping calendar features, supporting operator-level insights.

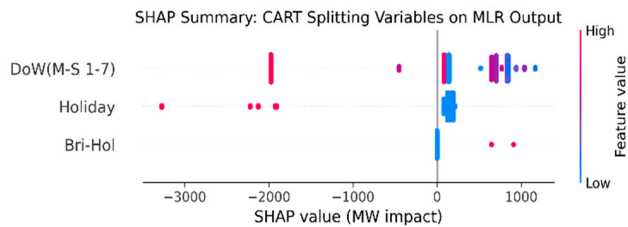


FIGURE 7. SHAP summary plot: Influence of CART splitting variables on MLR output via surrogate model.

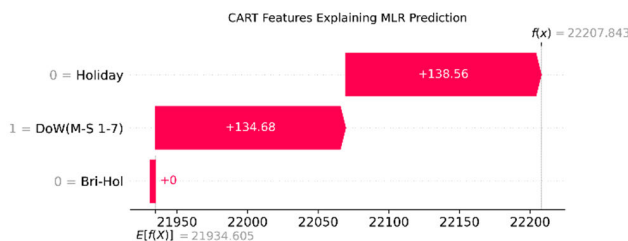


FIGURE 8. SHAP waterfall plot for a single instance for a normal day.

In terms of feature importance across periods, SHAP analysis confirmed that lagged load from the previous day consistently held the strongest influence, followed by

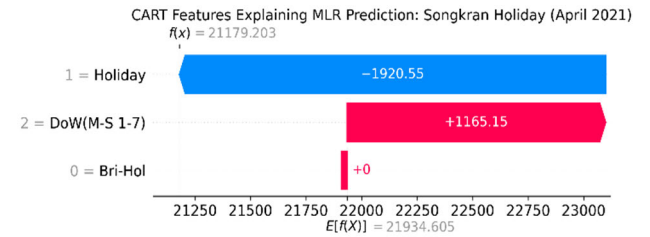


FIGURE 9. SHAP waterfall plot for a single instance for Songkran holiday.

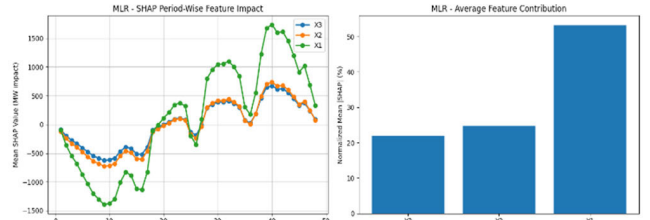


FIGURE 10. SHAP period-wise feature importance for MLR-based load forecasting.

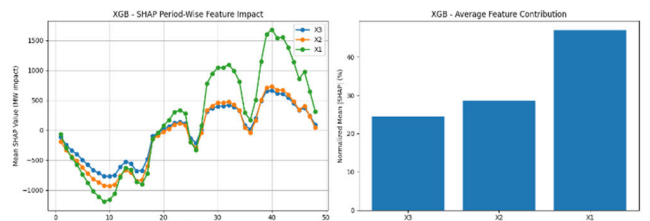


FIGURE 11. SHAP period-wise feature importance for XGBoost-based load forecasting.

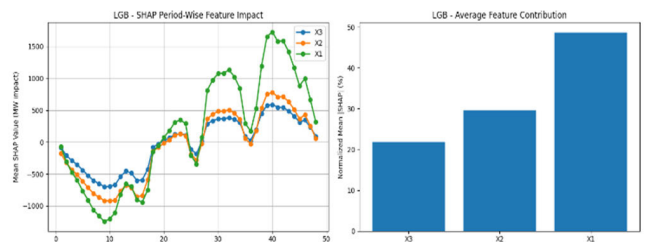


FIGURE 12. SHAP period-wise feature importance for LGB-Based load forecasting.

one-week and two-week lags. Figs. 10–12 compare this across MLR, XGBoost, and LightGBM. This confirms that short-term memory dominates electricity demand, reinforcing the framework’s reliance on recent consumption patterns.

By integrating SHAP at both calendar segmentation and lagged feature levels, the framework delivers interpretable forecasts without compromising performance. This dual-layer interpretability enables practical insights for utilities, especially in calendar-driven operational planning.

TABLE 10. Comparative evaluation of forecasting frameworks and published state-of-the-art techniques.

Ref No:	Calendar-Based Classification	Forecasting Model	Error Forecasting	XAI	Input Variables for Forecasting Models	MAPE (%)
[10]	Pruned CART	DBN	No	No	$(d-1)$, Hol, B_Hol, DoW	27.29
[30]	No	LSTM	No	No	$L_t(d-7)$, $L_{t-2}(d-1)$, $L_{t-1}(d-1)$, $L_t(d-1)$, $T_t(d-1)$, SI	4.93
[31]	No	Linear Regression - KF Adaptation	Partial (Adapted Coeffs)	No	$L_t^g(d-14)$, $L_t^g(d-7)$, $L_t^g(d-1)$	30.02
[21]	No	MLR	Yes	No	$L_t^g(d-14)$, $L_t^g(d-7)$, $L_t^g(d-1)$	5.77
[18], [29]	No	GAM	No	No	$L_t^g(d-7)$, $L_t^g(d-1)$	9.45
	CART	GAM	No	No	$L_t^g(d-7)$, $L_t^g(d-1)$	8.97
	CART	GAM-GPR-GLSAR	Yes	No	$L_t^g(d-14)$, $L_t^g(d-7)$, $L_t^g(d-1)$	0.90
	No	MLR	No	No	$L_t^g(d-14)$, $L_t^g(d-7)$, $L_t^g(d-1)$	6.71
	CART	MLR	No	No	$L_t^g(d-14)$, $L_t^g(d-7)$, $L_t^g(d-1)$	4.49
	CART	MLR-GPR	No	No	$L_t^g(d-14)$, $L_t^g(d-7)$, $L_t^g(d-1)$	3.24
	CART	MLR-GLSAR	Yes	Yes	$L_t^g(d-14)$, $L_t^g(d-7)$, $L_t^g(d-1)$	1.27
Proposed Model	CART	MLR-GPR-GLSAR	Yes	Yes	$L_t^g(d-14)$, $L_t^g(d-7)$, $L_t^g(d-1)$	0.72

H. BENCHMARK COMPARISON AND STATE-OF-THE-ART ANALYSIS

Table 10 presents a benchmarking analysis comparing the proposed framework with relevant literature in the domain of STLf. Early efforts such as [10] employed pruned CART classification coupled with DBNs, yielding a MAPE of 27.30%, highlighting the instability of deep architectures under low-resolution calendar inputs. Even in the same study, a CART-MLR setup achieved a modest 4.49%. More recent models, such as GAM in [18] and [29], reported a 3.78% MAPE without any explainability or residual error modeling.

A comparable LSTM-based method [30] utilized extensive weather and time-series lag inputs but lacked reported MAPE and explainability, limiting reproducibility and practical adoption. Manual calendar grouping strategies used in conventional MLR pipelines incorporated SHAP but not statistical error modeling, resulting in a higher MAPE (4.15%).

We also benchmark against the adaptive Linear Regression–Kalman Filter (KF) technique presented in [31], a method incorporated within the winning solution of the IEEE DataPort Competition on Day-Ahead Electricity Load Forecasting. This approach treats regression coefficients as time-varying states estimated via a Kalman Filter, with explanatory variables serving as observation matrices. Our implementation—restricted to the calendar and lagged load features used in this study and tuned for realistic process/observation noise—achieved an overall MAPE of 30.02%. While the full strategy in [31] leverages rich meteorological inputs, our reduced-feature replication provides a relevant baseline representing adaptive linear modelling.

Importantly, Table 10 also functions as an ablation analysis of the proposed forecasting framework. Each row corresponds to a progressively enhanced model variant, allowing the contribution of each module to be examined independently. The transition from MLR to CART+MLR quantifies the benefit of calendar-based segmentation; CART+MLR to CART+MLR–GPR highlights the importance of substituting GPR in sparse, holiday-related leaf nodes; and CART+MLR–GPR to CART+MLR–GPR–GLSAR demonstrates the substantial impact of correcting serially correlated residuals through GLSAR.

In contrast, the proposed hybrid achieves $\text{MAPE} = 0.72\%$ and $R^2 = 0.99$, a substantial improvement over state-of-the-art models, achieved without relying on exogenous weather variables. This demonstrates that error-corrected, explainable, and calendar-driven segmentation can outperform more complex deep architectures in both accuracy and interpretability.

I. COMPUTATIONAL COMPLEXITY AND RUNTIME PERFORMANCE

Computational efficiency is critical for real-time forecasting. The CART segmentation step required approximately 53 seconds to generate the tree-based calendar splits; as a one-time preprocessing task, its computational cost is negligible compared to repeated forecasting. Table 11 reports the runtime of all models across CART-derived segments. Statistical models (MLR, GAM) are complete within minutes, while boosted ensembles (LGB, XGB) require tens of minutes to about one hour. Deep learning approaches are more

computationally demanding, with NHITS finishing in under 12 minutes and CNN-LSTM requiring more than 3 hours.

TABLE 11. Comparative computational cost of forecasting models across all leaf nodes.

Model	Base Model_Run Time	GLSAR_Run Time	Total_Run Time
MLR	6m 44.8s	1m 29.4s	8m 74.2s
LGB	12m 29.3s	21.4 s	12m 50.7s
XGB	1h 5m 51.6s	19.3 s	1h 6m 10.9s
GAM	15m 14s	17.1 s	15m 31.1s
NHITS	11m 19.5s	23.7 s	11m 43.2s
CNN-LSTM	3h 24m 33.9s	1m 27.4s	3h 26m 1.3s
GPR	18m 33s	32.99s	19m 06s

For leaf nodes with limited training data, GPR completes in approximately 19 minutes, making it suitable for special-case scenarios such as holiday or bridging-day forecasts. Across all models, the GLSAR step adds only modest overhead (tens of seconds to a few minutes), confirming its scalability.

For interpretability, the total explainability pipeline required about 3 minutes and 9 seconds, which is acceptable for offline interpretability audits. Since SHAP is not needed in the operational pipeline, its runtime does not impact real-time forecasting performance.

All experiments were implemented in Python (scikit-learn, XGBoost, LightGBM, Stats models, PyGAM, TensorFlow) and executed on a Lenovo laptop (Intel 12th Gen Core i7-1255U CPU, 12 logical cores, 1.7 GHz, 16 GB RAM) running Windows 11 (64-bit). All experiments were executed on CPU without GPU acceleration; thus, reported runtimes represent conservative performance estimates suitable for real-time deployment.

J. SUMMARY OF KEY FINDINGS

- GLSAR integration substantially enhances accuracy and temporal stability across all model classes.
- CART-MLR-GPR-GLSAR provides the best trade-off between accuracy, interpretability, and computational efficiency.
- Holiday load forecasting benefits significantly from hybrid GPR correction, achieving robust performance under sparse conditions.
- SHAP-based interpretability enables both global and local insights into feature effects, supporting transparent decision-making for grid operators.
- Cross-dataset validation on France (2021–2023; 24 intervals/day) confirms strong generalization, with GLSAR reducing MAPE from 3.86% to 2.90% and increasing R^2 from 0.82 to 0.96 despite differing climatic and temporal structures.

Collectively, these results confirm that a calendar-aware, error-corrected, and explainable modeling approach can

TABLE 12. Summary of ablation findings.

Component	Contribution	Evidence
CART segmentation	Moderate boost	Reduces MAPE by ~55%, isolates homogeneous day types, stabilizes forecasting model behavior
GLSAR correction	Largest boost	Improves MAPE by 60–94%, shifts DW from ~1 → 2, reduces ACF, from 0.55–0.97 → 0.01–0.08, statistically significant ($p < 0.0001$)
GPR for sparse/holiday nodes	Essential for sparse data	Reduces MAPE from 23–46% → 2–9% in low-data nodes (Leaf 10–15), resolves instability where MLR fails
SHAP integration	Explainability + validation	Confirms feature influences (DoW, Holiday, Bridging-Hol, lagged loads) align with physical load behavior

achieve state-of-the-art short-term load forecasting performance while remaining interpretable and operationally practical.

IV. CONCLUSION

This study introduced a calendar-aware, weather-independent, and explainable day-ahead load forecasting (DALF) framework that integrates classification-based segmentation, statistical and machine learning models, residual autocorrelation correction, and post hoc interpretability. The proposed methodology employs a CART to partition daily load profiles based on calendar features, including day-of-week, holiday, and bridging holiday flags, thereby enabling segment-specific model training.

Within each segment, forecasting is conducted using lagged load inputs through MLR, with GPR selectively applied in data-sparse segments, particularly those affected by holidays. A GLSAR procedure is subsequently used to mitigate residual autocorrelation across daily intervals, significantly enhancing temporal consistency and model stability. To support model transparency, SHAP-based explainability is incorporated, offering both global and local insights into input feature contributions.

Empirical evaluations using real-world half-hourly load data from the Electricity Generating Authority of Thailand (EGAT) demonstrate that the proposed CART-MLR-GPR-GLSAR framework outperforms traditional, machine learning, and deep learning models. The integrated approach achieves an MAPE of 0.72% and an R^2 of 0.99 on daily load forecasts, excluding external weather variables. It establishes new benchmarks for both accuracy and interpretability in STLF under constrained input conditions. In particular, the incorporation of GLSAR led to consistent performance improvements across all baseline models, including XGBoost, LightGBM, GAM, and CNN-LSTM. Moreover, SHAP analysis verified the validity of the segmentation strategy and highlighted the dominant influence of recent historical load values on forecasting outcomes. Validation on the France 2021–2023 dataset further demonstrated the

generalizability of the proposed framework across different grid structures, seasonal patterns, and temporal resolutions.

The proposed framework is practical for operational deployment, especially in regions where weather data is unavailable or unreliable. Future research will explore the integration of adaptive online learning, probabilistic forecasting under uncertainty, and the extension of the methodology to multivariate spatiotemporal load prediction across multiple substations or regions. Additionally, Transformer-based architecture, such as the Temporal Fusion Transformer (TFT), will be investigated in future work. Their ability to capture long-range dependencies may further enhance forecasting accuracy and generalizability once appropriate computational resources and tuning budgets are allocated.

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